

Google Earth Engine (GEE) - Technical Reference

1. Integration Overview

HoubaraTracker integrates directly with the Google Earth Engine API to process satellite imagery (Sentinel-2, Landsat 8) in real-time to overlay environmental indices.

2. NDVI (Normalized Difference Vegetation Index)

Calculated as $(NIR - Red) / (NIR + Red)$. Used to assess the presence of live green vegetation.

3. SAVI (Soil Adjusted Vegetation Index)

Calculated as $((NIR - Red) / (NIR + Red + L)) * (1 + L)$ where $L = 0.5$. SAVI is specifically used in our desert contexts to minimize the influence of soil brightness.

4. LST (Land Surface Temperature)

Derived from Thermal Infrared sensor data, LST provides the temperature of the ground surface, helping to track the thermal stress on the birds.